

The empirical recurrence for OEIS A282882: recovering a conjecture that is not written down

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Abstract

OEIS A282882 records “Empirical recurrence of order 86”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 125$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 86, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 86 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282882 is “Number of nX5 0..1 arrays with no 1 equal to more than one of its horizontal, vertical and antidiagonal neighbors, with the exception of exactly one element.”. It has offset 1 and begins

5, 122, 2426, 34314, 480995, 6296324, 80114311, 995928444, ...

The entry states:

Empirical recurrence of order 86 (see link above)

As of the “Last modified” line on the live entry (Feb 24 2017 07:47:38, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 653$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 125$ of the 653, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 125$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 250$ exact terms of the model returns order 86 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{86} c_i a(n-i),$$

c_1	2	c_{23}	−6408765802	c_{45}	4099436891982	c_{67}	34720570
c_2	121	c_{24}	5479959446	c_{46}	−2651385252652	c_{68}	−1819425
c_3	594	c_{25}	3611540498	c_{47}	1580273162512	c_{69}	−4139362
c_4	−2211	c_{26}	−24738203502	c_{48}	−905524532124	c_{70}	5425100
c_5	−41076	c_{27}	494023360876	c_{49}	483202739240	c_{71}	−3867998
c_6	−262018	c_{28}	−1873129669107	c_{50}	−194273848822	c_{72}	1620827
c_7	−1095804	c_{29}	4072475993600	c_{51}	28621730284	c_{73}	−571582
c_8	−3398839	c_{30}	−7422887000085	c_{52}	7976942400	c_{74}	181175
c_9	−7942116	c_{31}	12511561896526	c_{53}	−10084083424	c_{75}	−33846
c_{10}	−13039089	c_{32}	−18275617317213	c_{54}	26439014558	c_{76}	−9447
c_{11}	−10421632	c_{33}	23317888705516	c_{55}	−31244180418	c_{77}	16422
c_{12}	15388295	c_{34}	−27852528475965	c_{56}	22210496963	c_{78}	−7119
c_{13}	75569218	c_{35}	31018085464074	c_{57}	−13837756380	c_{79}	2500
c_{14}	139792040	c_{36}	−31018849905310	c_{58}	9954380048	c_{80}	−1756
c_{15}	93737798	c_{37}	29026080096304	c_{59}	−7158874548	c_{81}	486
c_{16}	−181885190	c_{38}	−26369885047073	c_{60}	3316668091	c_{82}	−71
c_{17}	−524949910	c_{39}	22162141555440	c_{61}	−461651188	c_{83}	64
c_{18}	−523415427	c_{40}	−17464173705608	c_{62}	−421304479	c_{84}	−10
c_{19}	535709316	c_{41}	13971259671982	c_{63}	504976806	c_{85}	0
c_{20}	2198680700	c_{42}	−10855138061147	c_{64}	−434068818	c_{86}	−1
c_{21}	2748310694	c_{43}	7852134656388	c_{65}	280091194		
c_{22}	−8037804912	c_{44}	−5715184748963	c_{66}	−127989180		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 86, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $86 + 4$ terms removed returns order 86 as well, so $\deg q_{\min} = 86 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $86 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 86 that this sequence satisfies — and that family turns out to have exactly one member.

References

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